

Supplemental Information for “Does provable absence of barren plateaus imply classical simulability?”

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Supp. Info. A: Barren plateau-free models, polynomial subspaces, and simulation algorithms

We now present an analysis for each architecture and technique in Tables 1 and 2. In all cases we begin by defining the problem class of interest, reviewing the proof of absence of barren plateaus (i.e., determining the measurement operators and initial states leading to non-exponential concentration), and using this information to find the polynomially-large subspace. We then present tomographic techniques that can allow us to obtain the projection of the initial state onto the relevant subspace, and some appropriate simulation algorithms that can be used to classically compute the loss. In particular, here we show that that loss functions in Tables 1 and 2 are in QESIM.

1. Small dynamical Lie algebras

Since several examples in Tables 1 and 2 fall inside of this category, we find it convenient to first present the abstract case of circuits with small dynamical Lie algebras.

We begin by (very briefly) reviewing some preliminary concepts that will be useful throughout the rest of the Appendices. We note that the goal of this perspective article is not to delve on mathematical details. As such, we will simply define the *dynamical Lie algebra* of a circuit, and also what an *equivariant* operator is. We refer the reader to the cited articles for further details.

First, let us assume that the parametrized quantum circuit takes the form

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L e^{-i\theta_l H_l}, \quad (\text{A1})$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots)$ are the trainable parameters, and H_l Hermitian operators taken from some set of generators $\mathcal{G}$. We note that Eq. (A1) can also capture an ansatz with fixed, un-parametrized, layers by setting some of the parameters

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to be constant. It is known that one can measure the potential expressivity of a circuit by analyzing its dynamical Lie algebra [1–6]

$$\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}, \quad (\text{A2})$$

where $\langle \cdot \rangle_{\text{Lie}}$ denotes the Lie closure. As such, $\mathfrak{g} \subseteq \mathfrak{u}(2^n)$ is the real vector space spanned by the nested commutators of the elements in $i\mathcal{G}$. Here $\mathfrak{u}(2^n)$ denotes the unitary algebra on the Hilbert space of n qubits. The importance of $\mathfrak{g}$ stems from the fact that any unitary produced by the circuit (for any number of layers L , and for any set of parameters θ) is contained in the subgroup generated by $\mathfrak{g}$, i.e., $U(\theta) \in e^{\mathfrak{g}}$.

Recently, it was realized that too expressive circuits (where $\mathfrak{g}$ is exponentially large) are prone to exhibit barren plateaus [2, 7–9]. As such, a significant amount of effort has been dedicated to understanding how the expressivity of the circuit could be reduced. One of the most promising approaches is to create circuits that respect a given symmetry group S [10–16]. This led to the concept of *equivariant* circuits, where all the circuit’s generators commute with the all the elements in S . More generally, we say that any operator is equivariant if it commutes with all the elements in S .

With the previous, definitions in mind, we denote by $\mathcal{C}_{\text{polyg}}$ the class of all instances where the circuit’s dynamical Lie algebra $\mathfrak{g}$ is simple and of polynomial dimension, i.e., $\dim(\mathfrak{g}) \in \mathcal{O}(\text{poly}(n))$. We take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and O some Hermitian operator. All the gates in the circuit are parametrized, and every parameter is sampled uniformly at random. We further assume that either ρ or O are in $i\mathfrak{g}$.

Recently, an exact formula for deep versions of the circuits in $\mathcal{C}_{\text{polyg}}$ was simultaneously obtained in Refs. [8, 9]. In particular, these works show that the variance is inversely proportional to $\dim(\mathfrak{g})$, as conjectured in Ref. [2]. Moreover, if the projection of ρ and O in $i\mathfrak{g}$ is at most polynomially vanishing with the number of qubits, then it is guaranteed that a barren plateau will not appear. That is,

$$\mathcal{C}_{\text{polyg}} \subset \overline{\text{BP}}, \text{ if the projection of } \rho \text{ or } O \text{ in } i\mathfrak{g} \text{ is in } \Omega(1/\text{poly}(n)). \quad (\text{A3})$$

Notably, the relevant subspace in this case is the DLA itself. This follows from the fact a Lie algebra is closed under the adjoint action of its associated Lie group, i.e., if $g \in \mathfrak{g}$, then $UgU^\dagger \in \mathfrak{g}$ for all $U \in e^{\mathfrak{g}}$. Consequently, if O (or ρ) is in $i\mathfrak{g}$, the adjoint action of the group on it will result in an operator that lives inside the polynomially sized subspace. Hence,

$$\mathcal{C}_{\text{polyg}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A4})$$

Next, one needs to obtain the projection of ρ and O into the dynamical Lie algebra (denoted as $\rho_{\mathfrak{g}}$ and $O_{\mathfrak{g}}$, respectively), which requires computing the inner product of the initial state and the measurement operator with the elements of a basis of $i\mathfrak{g}$. Such computation will generally be case dependent, but in all cases considered we have always found an efficient tomographic procedure (see below for more information). From here, we can employ a recently introduced classical simulation algorithm called $\mathfrak{g}$ -sim [17, 18], which allows us to (back-)propagate the elements of the dynamical Lie algebra with unitaries of the Lie group. For this algorithm to work, one needs to compute the structure constants of the algebra. Importantly, computing these constants takes a time that scales polynomially in $\dim(\mathfrak{g})$. Thus, this suffices to show that

$$\mathcal{C}_{\text{polyg}} \subset \text{QESIM}. \quad (\text{A5})$$

We remark here that even if obtaining the structure constants takes polynomial time, such computational cost may still be prohibitively large [19]. It is then possible that polynomial speed-ups could be obtained by running the quantum circuit in the quantum computer instead.

Here we find it important to note that we have assumed above that the Lie algebra is simple. However, $\mathfrak{g}$ will generally decompose as a direct sum of simple Lie algebras (plus an abelian component) as $\mathfrak{g} = \oplus_{\eta} \mathfrak{g}_{\eta}$. Here, barren plateaus can be avoided (and thus simulations are enabled) if at least one of the simple components is polynomially large, and ρ and O have large projections therein.

As we will see below, the following examples from Tables 1 and 2 actually avoid barren plateaus by having a small Lie algebra: $U(1)$ -equivariant; and S_n -equivariant.

a. $U(1)$ -equivariant circuit

First, let us note that by $U(1)$ -equivariant circuits, we mean circuits that preserve the Hamming weight of a computational basis state. Hamming weight-preserving circuits have seen use for quantum chemistry, condensed matter, nearest centroid classification, portfolio optimization and hedging [20–23].

We define as $\mathcal{C}_{U(1)\text{equiv}}$ the class of all instances where the circuit is $U(1)$ -equivariant, and is composed of Hamming weight preserving local gates. We take ρ to be an n -qubit pure state with fixed Hamming weight k preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and O some $U(1)$ -equivariant operator. All the gates in the circuit are parametrized, and every parameter is sampled uniformly at random.

Several proofs of absence of barren plateaus for $U(1)$ -equivariant have been presented in the literature [2, 23, 24]. Crucially, they all somewhat rely on the fact that the restriction of the dynamical Lie algebra into the subspace with k Hamming weight can be, at most, of dimension $\mathcal{O}(\binom{n}{k}^2)$, meaning that a necessary condition for the loss function to not exhibit barren plateaus is that the initial state's Hamming weight is not too large. Note that bitstrings with Hamming weight $n - k$ can trivially be mapped to bitstrings of Hamming weight k by flipping all bits, and, while our following statements equally apply to Hamming weight $n - k$, we will focus on the case of Hamming weight $k \leq n/2$. As such, one finds

$$\mathcal{C}_{U(1)\text{equiv}} \subset \overline{\text{BP}}, \text{ if the Hamming weight } k \text{ of } \rho \text{ is such that } k \in \mathcal{O}(1). \quad (\text{A6})$$

The previous description allows us to readily identify the polynomially small subspace. Namely, since the initial state has fixed Hamming weight k and since the unitary and measurement preserve such Hamming weight, then all of the dynamics occur in the Hilbert's subspace of states with Hamming weight k . Moreover, we know that the dimension of such subspace is $\binom{n}{k}$, which is in $\mathcal{O}(\text{poly}(n))$ provided that $k \in \mathcal{O}(1)$. Hence, $\forall \theta$ one has

$$\mathcal{C}_{U(1)\text{equiv}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A7})$$

From here, we can perform tomography of ρ and O in a basis of Hamming weight k states to obtain an efficiently storable description in the polynomial-sized subspace. Moreover, while we can use $\mathfrak{g}$ -sim to simulate the loss functions in $\mathcal{C}_{U(1)\text{equiv}}$, here we can also obtain the action of $U(\theta)$ in the subspace by studying how each local gate acts on a computational basis state, a procedure that is highly efficient (see Ref. [25] for an example with Givens rotations). These ingredients then suffice to show that

$$\mathcal{C}_{U(1)\text{equiv}} \subset \text{QESIM}. \quad (\text{A8})$$

b. S_n -equivariant

Many problems in physics, machine learning, and quantum metrology have properties that remain invariant under any possible permutation of the n qubits in the system. As such, studying these problems variationally will require the use of parametrized quantum circuits that are equivariant with respect to S_n , the symmetric group acting on n qubits via qubit permutation [10].

We define as $\mathcal{C}_{S_n\text{equiv}}$ the class of all instances where the circuit is composed of S_n -equivariant layers of one and two-qubit gates. We take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and O some S_n -equivariant operator. All the gates in the circuit are parametrized, and every parameter is sampled uniformly at random.

In Ref. [6] the authors studied the dynamical Lie algebra associated to the unitaries used in $\mathcal{C}_{S_n\text{equiv}}$ and determined that $\dim(\mathfrak{g}) \in \mathcal{O}(n^3)$. In particular, the absence of barren plateaus for this problem class was studied and proved in Ref. [26], where, as expected, the authors found that

$$\mathcal{C}_{S_n\text{equiv}} \subset \overline{\text{BP}}, \text{ if the projection of } \rho \text{ in } i\mathfrak{g} \text{ is in } \Omega(1/\text{poly}(n)). \quad (\text{A9})$$

Since the measurement operator is in the algebra, and since the unitary cannot take O outside of the algebra, we find that the subspace is proper for all θ and thus

$$\mathcal{C}_{S_n\text{equiv}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A10})$$

As discussed above, the simulation can be performed via **g**-sim provided that we can compute the components of the initial state and measurement operator in the Lie algebra. Notably, this information can be obtained via the tomographic procedure of permutation invariant shadows introduced in Ref. [27], indicating that

$$\mathcal{C}_{\text{Snequiv}} \subset \text{QESIM}. \quad (\text{A11})$$

2. Matchgate circuit and small modules

Matchgate circuits, also known as fermionic linear optics circuits, are circuits whose gates can be mapped to free fermion evolution [28]. Their parameterized version [23, 29–32] has received considerable attention in variational quantum computing as a test bed for benchmarking different ideas [33]. Crucially, the study of quantum resources and simulability of matchgate circuits [34–37] has been recently related to the theory of barren plateaus [38].

We denote by $\mathcal{C}_{\text{match}}$ the class of all instances where the circuit is a parametrized matchgate circuit composed of parametrized rotations about the z axis and parametrized entangling XX gates on neighboring qubits. We take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and O some Pauli operator. All of the gates in the circuit are parametrized, and every parameter is sampled uniformly at random.

The presence and absence of barren plateaus in parametrized matchgate circuits was originally empirically studied in Refs. [2, 39], with a recent full characterization presented in Ref. [38]. In particular, it is well known that the dimension of these circuits' dynamical Lie algebra scales as $\mathcal{O}(n^2)$. This already means that, if O or ρ are in $i\mathfrak{g}$, we know that barren plateaus can be avoided, as the algebra itself is the relevant polynomially small subspace. However, the underlying algebraic properties of this model allow us to precisely characterize the loss function variance for *any* initial state and measurement operator. Here, the key realization is that under the adjoint action of the circuit's unitary, operator space $\mathcal{B}$ decomposes into group-modules as $\mathcal{B} = \bigoplus_{\eta=1}^{2n} \mathcal{B}_\eta$, where the operators in $\mathcal{B}_\eta$ can be exactly characterized as the product of η Majorana operators, and thus is of dimension $\binom{2n}{\eta}$. Hence, if $\eta \in \mathcal{O}(1)$ for $\eta \leq n$, or if $2n - \eta \in \mathcal{O}(1)$ for $\eta > n$, one obtains $\dim(\mathcal{B}_\eta) \in \mathcal{O}(\text{poly}(n))$. In what follows, we will assume for simplicity $\eta \leq n$. Note that these subspaces are proper for all θ . Assuming that O is in some $\mathcal{B}_\eta$ we get

$$\mathcal{C}_{\text{match}} \subset \overline{\text{BP}}, \text{ if the projection of } \rho \text{ in } \mathcal{B}_\eta \in \Omega(1/\text{poly}(n)) \text{ and } \eta \in \mathcal{O}(1). \quad (\text{A12})$$

Here, identifying the small subspace is straightforward as the group modules are, by definition, closed under the adjoint action of the group. This allows us to say that if $\eta \in \mathcal{O}(1)$, then

$$\mathcal{C}_{\text{match}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A13})$$

Next, we need to obtain the component of ρ into the module $\mathcal{B}_\eta$ (note that since O is a Pauli in $\mathcal{B}_\eta$, we already have all the information we need from the measurement operator). Here, one simply needs to prepare ρ and measure the expectation value of all the Pauli operators that can be expressed as a product of η distinct Majoranas. Such a procedure is efficient if $\eta \in \mathcal{O}(1)$. Then, we can obtain the adjoint action of the unitary in each module by computing the commutator graphs [38] (or generalized structure constants) and using **g**-sim. The classical cost here scales polynomially with the dimension of the module, and is therefore polynomial in the number of qubits. Hence, we have

$$\mathcal{C}_{\text{match}} \subset \text{QESIM}. \quad (\text{A14})$$

3. Shallow hardware efficient ansatz

Since the shallow hardware efficient ansatz was presented in the main text, we only recap it briefly here for completeness.

First, we recall that as shown in Ref. [40], and also at the end of this Supplemental Information, one can prove that

$$\mathcal{C}_{\text{shallowHEA}} \subset \overline{\text{BP}}, \text{ if } \rho \text{ follows area law and } O \text{ is local}. \quad (\text{A15})$$

Moreover, since the measurement operator is local, the bounded light cone structure of a shallow hardware efficient ansatz means that the adjoint action of the circuit can only map the measurement operator to Paulis acting on at most $2L \in \mathcal{O}(\log(n))$ qubits. There are $\mathcal{O}(\text{poly}(n))$ such operators and so the resulting subspace is a proper polynomial-sized subspace for any θ . Hence

$$\mathcal{C}_{\text{shallowHEA}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A16})$$

To perform an efficient classical simulation of the loss we can obtain the adjoint action of the circuit in the subspace by ‘dropping’ from $U(\theta)$ all the gates that are outside of the measurement’s backwards light cone. Then, the final step is to determine ρ_λ . If ρ has a known classical representation, then one can classically find its projection onto the subspace by computing (e.g., via pen and paper or using classical computing power) the expectation value for all operators $P_j \in \mathcal{B}_\lambda$. However, if ρ is some state obtained at the output of a give circuit, these expectation values cannot generally be obtained classically. However, with access to a quantum computer one can efficiently estimate ρ_λ using local tomography of the reduced states. Notably instead of performing full tomography, one can also perform standard Pauli classical shadow tomography [41, 42] That is, we prepare ρ , and implement randomly selected single-qubit Clifford gates on each qubit, and measure them in the computational basis. Since we care about estimating the expectation value of polynomially many local operators, this tomographic procedure is efficient. As such, we find

$$\mathcal{C}_{\text{shallowHEA}} \subset \text{QESIM}. \quad (\text{A17})$$

It follows that no problem based on shallow hardware efficient ansätze with a local measurement can be outside of QESIM. We note that this result was originally reported in Ref. [43], with similar ideas explored concurrently in the context of quantum process learning in Ref. [44].

a. Quantum generative modeling

Quantum generative modeling is a sub-field of quantum machine learning where the goal is to learn an unknown probability distribution underlying a classical dataset, with the hope that the trained model can be utilized to generate new data samples that closely match the original data [45–50]. While several schemes exist for quantum generative modeling, we will here focus on quantum circuit generative models, which include quantum circuit Born machines (QCBMs) and quantum generative adversarial networks (QGANs). In these models, one prepares a parametrized quantum state that encodes the desired distribution over computational basis measurements. It has been shown that these models can avoid barren plateaus by using losses that contain significant local contributions, which, in conjunction with a shallow hardware efficient ansatz, guarantees non-exponential concentration [51, 52].

We define as $\mathcal{C}_{\text{gen}}$ the class of all instances where the circuit is a shallow one-dimensional hardware efficient ansatz composed of $L \in \mathcal{O}(\log(n))$ layers of two-qubit gates acting on neighboring-qubits in a brick-layered structure. Moreover, we take ρ to be the all-zero state, and O to be a weighted sum of projectors onto computational basis states. Here, we further assume that all the gates in the circuit are parametrized, and every parameter is sampled uniformly at random.

The loss functions in $\mathcal{C}_{\text{gen}}$ can be simulated using the fact that the whole circuit can be efficiently represented via tensor networks [53]. In particular, the all-zero state, when evolved by a shallow one-dimensional hardware efficient ansatz up to $\mathcal{O}(\log(n))$ depth, admits an efficient matrix product state [54] description (a proper subspace) for all θ . That is,

$$\mathcal{C}_{\text{gen}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A18})$$

Moreover, since there is no need for tomographic measurements, and since sampling from the output distribution of an MPS is efficient [55, 56], one has

$$\mathcal{C}_{\text{gen}} \subset \text{CSIM}. \quad (\text{A19})$$

4. Generic shallow locally circuits

Here, by generic shallow locally circuits we refer to circuits where local two-qubit gates act according to some arbitrary topology, and where, there is a path that connects the input state with the measurement operator that crosses, at most,

$\mathcal{O}(\log(n))$ such gates. This architecture encompasses hardware efficient ansatz in one-dimension, quantum convolutional neural networks, but also more general circuits where gates can act according to a generic topology (i.e, 2D and beyond). Importantly, the fact that the gate connectivity is arbitrary means that light-cone arguments such as ones previously presented for the shallow hardware efficient ansatz do not hold.

Let us then define as $\mathcal{C}_{\text{local}}$ the class of all instances where the parameterized quantum circuit is built from local gates acting on pairs of qubits according to some arbitrary topology. We then take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and the measurement operator O to be a local Pauli acting on a qubit at the circuit's output. We assume that the parameters are sampled such that each gate forms an independent 2-design over $SU(4)$.

The proof of absence of barren plateaus for these architectures can be derived with the results in [57, 58], but for completeness, we also present a proof at the end of this Supplemental Information (Appendix C). As shown therein, if the initial state is pure and satisfies an area law of entanglement, we find

$$\mathcal{C}_{\text{local}} \subset \overline{\text{BP}}, \text{ if } \rho \text{ follows an area law of entanglement.} \quad (\text{A20})$$

Moreover, the scrambling nature of the circuit effectively suppresses the contribution of high weight Paulis and the dynamics occur in a subspace composed of Paulis with $\mathcal{O}(1)$ -bodyness qubits, so that

$$\mathcal{C}_{\text{local}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A21})$$

For a rigorous proof see Ref. [59]. Then, the component of the initial state therein can be obtained via classical shadows, and the simulation follows as a direct application of the algorithm in [59], showing that

$$\mathcal{C}_{\text{local}} \subset \text{QESIM}. \quad (\text{A22})$$

5. Small angle initialization

A recurring theme in this and the following sections will be how to classically simulate the loss function values for effective polynomial subspaces for problem classes in QESIM without relying on strong symmetries as in Secs. A 1 & A 2 or small entanglement light cones such as in Sec. A 3. Broadly speaking, one can either *forward* propagate the initial state ρ (or measurements of it) through the circuit, or *backward* propagate the measurement operator O in order to estimate the loss $\ell_{\theta}(\rho, O)$.

If no tomography of the initial state is needed, i.e., if an efficient MPS or MPS representation is known, the approach of forward propagation can be especially efficient in practice. For quantum circuits with local entangling gates and at most $\mathcal{O}(\log(n))$ layers, the bond dimension of the tensor network will not grow to be exponentially large [53], but even beyond log-depth circuits, high quality approximation with polynomial bond dimension may be possible. If a polynomial number of measurements of a non-trivial initial state are given, e.g., using standard Pauli classical shadows [41], we notice that we can represent each measurement as an efficient matrix product operator (MPO) [60, 61] with bond dimension $\chi = 1$. The loss function $\ell_{\theta}(\rho, O)$ then becomes an average over all measurements from the quantum computer and their corresponding forward propagated MPOs.

On the flip side, one can also backward propagate the target operator through the quantum circuit, which may be particularly efficient in practice if O is relatively simple in some basis (e.g., the Pauli basis) and one has to collect measurements from a non-trivial initial state ρ . This can again be done using tensor networks in the form of MPOs or using methods, such as Pauli Propagation [62, 63], which work using the Pauli transfer matrix formalism [64–66].

The above techniques could be used for the classical simulation of both small angle initializations (discussed in this section) and QCNNs (discussed in the next section). Small angle initializations have been an increasingly popular strategy aimed at replacing the most commonly used random angle initialization techniques. Here we will be focusing on two of them: Hamiltonian variational ansatz with small angles and Gaussian initialization in deep variational quantum circuits.

a. *Hamiltonian variational ansatz*

The Hamiltonian variational ansatz was proposed in Ref. [67] as a problem-inspired circuit to find the ground state of a given target Hamiltonian H . Here, one splits H into a sum of terms $H = \sum_{i=1}^M \alpha_i h_i$, and builds the circuit as $U(\boldsymbol{\theta}) = \prod_{l=1}^L \prod_{i=1}^M e^{-i\theta_{l,i} h_i}$, leading to a circuit that somewhat resembles a Trotterized evolution.

Let us define as $\mathcal{C}_{\text{HVA}}$ the class of all instances where the parameterized quantum circuit is built is a Hamiltonian variational ansatz for a given H , which can be expressed as $H = \sum_{i=1}^M \alpha_i h_i$. We assume that each h_i is a k -local operator. We then take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state and the observable to be local, and the measurement operator O to be a local Pauli. We assume that the parameters are sampled in a region around zero whose width scales as $\mathcal{O}(1/(nML))$.

The proof of absence of barren plateaus is given in Ref. [68], where such parametrized quantum circuit is shown to be equivalent to a time evolution driven by a single k -local Hamiltonian. In this new framework, the loss can be shown to avoid barren plateaus if the initial state is pure and satisfies an area law of entanglement. Hence, we have

$$\mathcal{C}_{\text{HVA}} \subset \overline{\text{BP}}, \text{ if } \rho \text{ follows an area law of entanglement.} \quad (\text{A23})$$

Given that $U(\boldsymbol{\theta})$ implements a short time evolution, the adjoint action of $U(\boldsymbol{\theta})$ on O lives, with high probability for $\boldsymbol{\theta} \sim \mathcal{P}$, in an effective polynomially-sized subspace of operators acting on $\mathcal{O}(1)$ qubits (see the proof of Theorem 4 in Ref [69] for a detailed proof). Thus, one has

$$\mathcal{C}_{\text{HVA}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A24})$$

Since the operators in the small subspace are all acting on at most $\mathcal{O}(1)$ qubits, we can utilize standard Pauli classical shadows on the initial state [41]. Then, since the parametrized quantum circuit can effectively be expressed as $U(\boldsymbol{\theta}) \simeq e^{-iK\tau}$, where K is a k -local Hamiltonian ($k \in \mathcal{O}(1)$) and $\tau \in \mathcal{O}(1/n)$, we know that the entanglement generated by the circuit is bounded, and we can efficiently compute the loss function via tensor networks or Pauli Propagation methods (with time and sample complexity guarantees provided by Theorems 2 and 4 in Ref. [69]). Hence,

$$\mathcal{C}_{\text{HVA}} \subset \text{QESIM}. \quad (\text{A25})$$

b. *Gaussian initialization in deep variational quantum circuits*

Next, let us define as $\mathcal{C}_{\text{GI}}$ the class of all instances where the parametrized quantum circuit is built employing L layers of single qubit parametrized rotations followed by an entangling layer of controlled Z gates. We take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state and the observable to be local, and the measurement operator O to be a local Pauli. We assume that the parameters are sampled according to a Gaussian distribution centered around zero and with variance in $\mathcal{O}(1/L)$.

The proof of absence of barren plateaus for this initialization technique was presented in Ref. [70], where the authors find that exponential concentration is avoided if ρ satisfies an area law of entanglement. Therefore, one has

$$\mathcal{C}_{\text{GI}} \subset \overline{\text{BP}}, \text{ if } \rho \text{ follows an area law of entanglement.} \quad (\text{A26})$$

When the parameters are sampled from a Gaussian distribution whose width decreases with the circuit depth, then the adjoint action of the unitary on the measurements operator lives, with high probability for $\boldsymbol{\theta} \sim \mathcal{P}$, in a low magic subspace. Concretely, as proven in Ref. [69], it belongs to an effective subspace composed of operators acting on $\mathcal{O}(\text{poly}(n))$ qubits that can be efficiently classically identified using the small-angle Pauli Propagation algorithm. It follows that

$$\mathcal{C}_{\text{GI}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A27})$$

This trade-off between the range of parameter values and circuit depth renders the quantum circuit effectively simulated with low magic simulation algorithms [69, 71]. Again we might need to perform Pauli measurements on the initial state ρ . Putting these results together leads to

$$\mathcal{C}_{\text{GI}} \subset \text{QESIM}. \quad (\text{A28})$$

We note that Ref. [72] also proves a trainability guarantee for small angle initialization in an ansatz composed of alternating controlled Z layers and single qubit rotations. One can show that this is classically simulable by a similar argument to the one above.

6. Quantum convolutional neural networks

Quantum convolutional neural networks (QCNNs) are a type of circuit architecture introduced in Ref. [73] which has seen widespread use in supervised classification problems on both classical and quantum data [73–77]. A QCNN is composed of parametrized gates acting on nearest neighbors which are interleaved with pooling layers where qubits are either measured or traced-out. Due to the way this architecture is constructed, its depth always scales logarithmically with the number of qubits, and the measurement operator is always local.

a. Standard QCNN with tracing-out pooling layers

Let us define as $\mathcal{C}_{\text{QCNN}}$ the class of all instances where the circuit is a QCNN composed of layers of two-qubit gates acting on nearest neighbors which are interleaved with pooling layers where half of the qubits are traced-out. We take ρ to be an n -qubit state preparable by a circuit with $\mathcal{O}(\text{poly}(n))$ gates when acting on the all-zero state, and O some Pauli operator acting on the last two qubits of the QCNN. All of the two-qubit gates in the circuit are parametrized, and every parameter is sampled uniformly at random. We assume that randomly sampling the parameters leads to all of the gates forming independent local 2-designs.

The proof of absence of barren plateaus for QCNNs can be found in Ref. [78], but can also be derived with the tools presented at the end of this Supplemental Information (Appendix C). Therein, the authors showed that if all of the two-qubit gates in the circuit form independent local 2-designs, then the loss function will not exhibit a barren plateau provided that the reduced states of ρ onto any pair of qubits are not exponentially close to the identity (e.g., if the initial state follows an area law of entanglement). Hence, we can find that

$$\mathcal{C}_{\text{QCNN}} \subset \overline{\text{BP}}, \text{ if } \rho \text{ follows an area law of entanglement.}$$

With the previous, let us try to identify what are the relevant polynomial-sized subspaces under the QCNN’s adjoint action. First, note that there always exists a set of parameters θ_j such that the QCNN is able to transform the local Pauli operator O into *any* other Pauli P_j . That is,

$$\forall P_j, \text{ then } \exists \theta_j \text{ such that } \frac{1}{d} \langle U^\dagger(\theta_j) O U(\theta_j), P_j \rangle = 1. \quad (\text{A29})$$

While this equation indicates that the adjoint action allows us to reach exponentially many Pauli operators, this does not mean that the adjoint action will typically lead to exponentially large subspaces. Indeed, one can verify that, with high probability for $\theta \sim \mathcal{P}$, $U^\dagger(\theta) O U(\theta)$ will live in an effective polynomially large subspace whose basis is composed of Pauli operators acting on at most $\mathcal{O}(1)$ qubits [59, 79]. Thus, with high probability for some $\theta \sim \mathcal{P}$, the QCNNs adjoint action over the measurement leads to an effective polynomially-sized subspace, and so

$$\mathcal{C}_{\text{QCNN}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A30})$$

Given that the small subspace contains operators acting on a small number of qubits, we can again perform standard Pauli classical shadows on the initial state [41] to obtain a classical description. Then, since the circuit is composed of local gates, and contains at most $\mathcal{O}(\log(n))$ layers, we can simulate the loss via tensor networks or any of the techniques outlined at the beginning of Sec. A 5. High-fidelity approximations of $O(\theta)$ likely require polynomial MPO bond dimension, and algorithms such as Pauli Propagation could employ a simple truncation based on operator weight which has for example been used in Refs. [63, 66]. Thus, we find

$$\mathcal{C}_{\text{QCNN}} \subset \text{QESIM}. \quad (\text{A31})$$

Indeed, we note that the recent work of [79] has explicitly implemented the previous classical simulation method, effectively dequantizing the quantum convolutional neural network architecture.

b. *QCNN with post-selected pooling layers*

Next we consider a class of QCNNs where the loss function depends on the outcomes of the measurements in the pooling layers; we denote this class of QCNNs as $\mathcal{C}_{\text{pl-QCNN}}$. This setting is typically studied in the context of error-correction, where one wants to learn a circuit that optimizes recovery given some noise channel and logical state [73]. Here, the circuit is a QCNN composed of layers of two-qubit gates acting on nearest neighbors, interleaved with pooling layers where half of the qubits are measured and used to control single-qubit unitaries. While simulation is only available with high probability for the case of $\mathcal{C}_{\text{QCNN}}$, we claim one can simulate those in $\mathcal{C}_{\text{pl-QCNN}}$ for all parameter values; that is,

$$\mathcal{C}_{\text{pl-QCNN}} \subset \text{QESIM}. \quad (\text{A32})$$

We assume there are p parameterized gates, each a 2-qubit unitary; we denote the unitary implemented by the QCNN as U_{θ} , where $\theta \in \mathbb{R}^p$. In the post-selected setting the loss is some function of the expectation values of observables of the form:

$$O = \sum_{\mathbf{m} \in \mathcal{M} \subseteq \{0,1\}^{n-k}} O_{m_1, \dots, m_{n-k}} \otimes |\mathbf{m}\rangle \langle \mathbf{m}| \quad (\text{A33})$$

for some subset $\mathcal{M} \subseteq \{0,1\}^{n-k}$ and k -local $O_{m_1, \dots, m_{n-k}}$ (assumed to have bounded Hilbert–Schmidt norm). A typical example of this setting is when a QCNN is used to learn the decoder of a quantum error correcting code where, given a circuit V encoding a logical k -qubit state ρ_L , one maximizes the probability of correcting t errors after an error channel $\mathcal{N}$ is applied [73]:

$$f(\theta) = \sum_{\substack{\mathbf{m} \in \{0,1\}^n: \\ \|\mathbf{m}\|_1 \leq t}} \text{Tr} \left(\rho_L \otimes |\mathbf{m}\rangle \langle \mathbf{m}| U_{\theta} \mathcal{N} \left(V \left(\rho_L \otimes |0\rangle \langle 0|^{n-k} \right) V^{\dagger} \right) U_{\theta}^{\dagger} \right). \quad (\text{A34})$$

Here, the input to the QCNN is the n -qubit state $\rho := \mathcal{N} \left(V \left(\rho_L \otimes |0\rangle \langle 0|^{n-k} \right) V^{\dagger} \right)$, $O_{m_1, \dots, m_{n-k}} = \rho_L$, and (for constant t) $|\mathcal{M}| = \mathcal{O}(n^t)$.

While QCNNs are defined via real parameters—i.e., their parameters are of infinite precision—by a net argument [44, 74] we can assume that each of the p parameterized gates is drawn from a finite set $S_{p,\epsilon}$ of 2-qubit gates up to incurring a total error ϵ in diamond distance. More concretely, Theorem 1 of the Supplemental Information of Ref. [74] states that such a net of cardinality:

$$S_{p,\epsilon} = \mathcal{O} \left(\text{poly} \left(\frac{p}{\epsilon} \right) \right) \quad (\text{A35})$$

exists. We thus assume this setting without loss of generality.

We now show Eq. (A32) via the following theorem.

Theorem 1 (Classical simulability of QCNNs). *Fix $\epsilon, \delta = \Theta(1)$. Consider a QCNN with p parameterized 2-qubit gates drawn from $S_{p,\epsilon}$, and consider:*

$$\text{Tr} (OU\rho U^{\dagger}), \quad (\text{A36})$$

where O is as in Eq. (A33), U is a unitary implementing the QCNN parameterized by p elements of $S_{p,\epsilon}$, and ρ is an n -qubit state. Using

$$N = \mathcal{O} \left(\frac{|\mathcal{M}|^2}{\epsilon^2} \left(p \log \left(\frac{p}{\epsilon} \right) + \log(|\mathcal{M}|) + \log(\delta^{-1}) \right) \right) \quad (\text{A37})$$

copies of ρ , with probability at least $1 - \delta$, one can construct a classical shadow representation of ρ that allows one to estimate Eq. (A36) to additive error ϵ over all parameter settings. This classical shadow representation can be constructed with no prior knowledge of U or O .

Proof. Consider a classical shadows description of ρ using the random Clifford basis protocol of Ref. [41], built using

$$N = \frac{300 |\mathcal{M}|^2 \|O_{\max}\|_{\text{F}}^2}{\epsilon^2} (p \ln(S_{p,\epsilon}) + \ln(|\mathcal{M}|) + \log(2\delta^{-1})) \quad (\text{A38})$$

copies of ρ . Here, $\|O_{\max}\|_{\text{F}}$ is the maximal Hilbert–Schmidt norm of the $O_{m_1, \dots, m_{n-k}}$, assumed to be $\mathcal{O}(1)$. By Ref. [41], this classical shadows description of ρ is of size $\text{poly}(N)$. Theorem S1 combined with Proposition S1 of [41] guarantees that, with probability at least $1 - \delta$, this classical shadows description suffices to estimate the expectation values of

$$M = S_{p,\epsilon}^p |\mathcal{M}| \quad (\text{A39})$$

observables with Hilbert–Schmidt norm upper-bounded by $\|O_{\max}\|_{\text{F}}$ each to additive error $\frac{\epsilon}{|\mathcal{M}|}$ in classical time $\text{poly}(N)$. All that remains is demonstrating that there are M observables being measured with Hilbert–Schmidt norm upper-bounded by $\|O_{\max}\|_{\text{F}}$ by this QCNN.

Now consider the set of $|\mathcal{M}|$ observables:

$$\{O_{m_1, \dots, m_{n-k}} \otimes |\mathbf{m}\rangle \langle \mathbf{m}| \}_{\mathbf{m} \in \mathcal{M}}. \quad (\text{A40})$$

The Hilbert–Schmidt norm of each $O_{m_1, \dots, m_{n-k}}$ is upper-bounded by $\|O_{\max}\|_{\text{F}} = \mathcal{O}(1)$ by definition. By additionally counting all $S_{p,\epsilon}^p$ choices for U , there are thus $S_{p,\epsilon}^p |\mathcal{M}|$ operators of the form:

$$U^\dagger (O_{m_1, \dots, m_{n-k}} \otimes |\mathbf{m}\rangle \langle \mathbf{m}|) U \quad (\text{A41})$$

with Hilbert–Schmidt norm upper-bounded by $\|O_{\max}\|_{\text{F}}$. The considered classical shadows description of ρ thus suffices to estimate the expectation value of each $U^\dagger (O_{m_1, \dots, m_{n-k}} \otimes |\mathbf{m}\rangle \langle \mathbf{m}|) U$ to additive error $\frac{\epsilon}{|\mathcal{M}|}$ with probability at least $1 - \delta$, and thus also of each $U^\dagger O U$ to additive error ϵ . $\square$

7. Non-unital noisy-circuits

Let us define as $\mathcal{C}_{\text{non-u}}$ the class of all instances where the circuit is composed of two-qubit gates on a one-dimensional lattice, interleaved by local (single-qubit) noise, with a final layer of single-qubit gates. Furthermore, we assume that the noise is non-unital and contractive [80]. We take ρ to be a generic n -qubit state, and the observable O to be local. We assume that all parameters are sampled such that each gate forms an independent 2-design.

As shown in [80], the effect of the non-unital noise is to effectively make the circuit shallow by “resetting” the qubits after each layer of parametrized gates. Indeed, as proved in Theorem 1 of [80], one can only consider the last $\mathcal{O}(\log(n))$ layers of the circuit, after which one can use light-cone arguments to prove that

$$\mathcal{C}_{\text{non-u}} \subset \overline{\text{BP}}. \quad (\text{A42})$$

Similarly, since the measurement is local, the same light-cone arguments can be used to show that the Heisenberg-evolved measurement operator will act on operators acting on at most $\mathcal{O}(\log(n))$ qubits. That is,

$$\mathcal{C}_{\text{non-u}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{A43})$$

Finally, by either using the classical simulation algorithms of Ref. [80] in the case of 1D circuits, or those in Refs. [59, 81] for arbitrary topologies, one obtains

$$\mathcal{C}_{\text{slocal}} \subset \text{QESIM}. \quad (\text{A44})$$

8. Parameterized Dynamic Circuit

Parameterised dynamic circuits make crucial use of *feedforward* operations. A feedforward operation consists of a measurement on a qubit followed by a conditional gate: If the measurement outcome is 0 then $U_0 = I$ is applied, if instead it is 1, then U_1 is applied where

$$U_1 = \begin{pmatrix} \cos \varphi e^{-i\phi} & -i \sin \varphi \\ -i \sin \varphi & \cos \varphi e^{i\phi} \end{pmatrix}. \quad (\text{A45})$$

Let us define $\mathcal{C}_{\text{dyn}}$ as the class of all instances where the circuit implements a channel $\mathcal{E}(\boldsymbol{\theta}) = \mathcal{U}_d(\boldsymbol{\theta}_d) \circ \dots \circ \mathcal{U}_1(\boldsymbol{\theta}_1)$, where each channel $\mathcal{U}_j(\boldsymbol{\theta}_j)$ is a completely positive, trace preserving map on n qubits. The operations in each $\mathcal{U}_j$ can be written as a composition of single-qubit and two-qubit operations in an arbitrary circuit topology. It is further assumed that:

1. Every component $\theta_i \in \boldsymbol{\theta}$ parameterizes only a single operation in $\mathcal{E}$ (i.e., the parameters are independent and uncorrelated).
2. $\mathcal{E}$ is locally scrambling, i.e., is invariant under single-qubit random gates from a unitary 2-design after every gate.
3. $\mathcal{E}$ has a constant worst-case feedforward distance. That is, we define f be the shortest path from a qubit measurement to a feedforward operation through the backwards light cone of the measurement and take $f \in O(1)$.
4. $\mathcal{E}$ has an average entangling power of α and average swapping power $\beta \geq 0$ [82].

Let O be a k -local Hamiltonian with $k \in \log(n)$ and let the initial state be arbitrary.

Then, as shown in Ref. [83], similarly to the case of non-unital noise, the effect of the resets is to effectively make the circuit shallow by “resetting” the qubits after each layer of parametrized gates. However, in contrast to non-unital noise, these resets can be implemented in a controlled manner. Indeed, as proven in Theorem 1 of [83], one finds that

$$\mathcal{C}_{\text{dyn}} \subset \overline{\text{BP}}. \quad (\text{A46})$$

Since these circuits are locally scrambling, the arguments of Ref. [84] also apply here, thus providing bounds on the approximation error by considering only the reduced subspace. In particular, the mid circuit measurements at the core of dynamic circuits can be viewed as controlled two qubit rotations followed by qubit resets and thus can be absorbed into the setting considered in Ref. [84]. Thus, we have

$$\mathcal{C}_{\text{dyn}} \subset \mathcal{C}_{\text{polySub}}, \quad (\text{A47})$$

and

$$\mathcal{C}_{\text{dyn}} \subset \text{QESIM}. \quad (\text{A48})$$

Supp. Info. B: Examples of non-concentrated but also non-simulable loss functions

In this section we showcase an example of a non-concentrated loss function which is not classically simulable. This example showcases that $\overline{\text{BP}} \neq \text{QESIM}$.

Let us begin by assuming that the parameterized quantum circuit takes the form,

$$U(\boldsymbol{\theta}) = U \left(\bigotimes_{i=1}^n e^{-i\theta_i X_i} \right), \quad (\text{B1})$$

where U is a fixed polynomial-size quantum circuit. More specifically, we take U to be a quantum circuit that implements a Boolean function $B(x) : \{0, 1\}^n \rightarrow \{-1, 1\}$ over the first output qubit that is classically hard to simulate for random

input bitstrings, even with advice strings [85]. Many cryptographic problems related to discrete logarithms are widely believed to satisfy these criteria because they are random self-reproducible. We take any one of them.

Next, let us focus on U and the corresponding Boolean function $B(x)$. The circuit U implements $B(x)$ in the following sense:

$$\|\text{Tr}[U|x\rangle\langle x|U^\dagger Z_1] - B(x)\| < 0.01, \quad (\text{B2})$$

for all $x \in \{0,1\}^n$, where Z_1 is the Pauli-Z operator acting on the first qubit. As previously stated, we will focus on the fact that $B(x)$ is classically hard to simulate for uniformly random input bitstrings $x \in \{0,1\}^n$, even given a polynomial-size advice string. Equivalently, this can be stated as saying that this class of Boolean functions $\{B(x)\}$ is *not* in BPP/poly. As data from quantum experiments are a restricted form of advice strings, we will use this observation to argue that a loss function that estimates $B(x)$ for x uniformly sampled from $\{0,1\}^n$ is not QESIM.

Specifically, let us consider the loss function

$$\ell_\theta(O) = \langle 0^n | \left(\bigotimes_{i=1}^n e^{i\theta_i X} \right) U^\dagger Z_1 U \left(\bigotimes_{i=1}^n e^{-i\theta_i X} \right) | 0^n \rangle, \quad (\text{B3})$$

where $O = Z_1$ and we have explicitly taken the initial state ρ to be the all-zero state. Let us further suppose that each θ_i is uniformly sampled from the *discrete* set $\{0, \pi/2, \pi, 3\pi/2, 2\pi\}$ such that the parametrized quantum circuit prepares a random computational basis state. That is, $(\bigotimes_{i=1}^n e^{-i\theta_i X}) | 0^n \rangle = |x\rangle$ where $|x\rangle$ is a random bitstring. We call this distribution $\mathcal{P}_{\text{disc}}$. We will then define $\mathcal{C}_{\text{Bool}}^{\text{disc}}$ as the class of all instances where the circuit is of the form in Eq. (B1) where U implements a classically hard Boolean function (even with advice strings) and $\theta \sim \mathcal{P}_{\text{disc}}$.

Since this circuit by assumption approximates $B(x)$ for all x , as stated in Eq. (B2), and estimating $\{B(x)\}$ for uniformly sampled x is not in BPP/poly, and data from quantum experiments are a form of advice string, it follows that $\mathcal{C}_{\text{Bool}}^{\text{disc}}$ is not in QESIM. Moreover, since U is a polynomial-size circuit, the problem is in QSIM. That is, we have

$$\mathcal{C}_{\text{Bool}}^{\text{disc}} \subset (\text{QSIM} \cap \overline{\text{CSIM}_{\text{QE}}}), \quad (\text{B4})$$

where $\overline{\text{CSIM}_{\text{QE}}}$ denotes the classes not in QESIM.

It remains to show that this problem does not exhibit a barren plateau. To show this we start by highlighting that $B(x)$ has a large variance over the input bitstrings. To see this note that if the variance is smaller than $1/100$, then for at least a $9/10$ fraction of the inputs, $B(x)$ must give the same output (the constants are not optimized here, this is just for illustrative purposes). As a result, the error of a trivial classical algorithm that always gives the same output, would only be wrong on at most $1/10$ of the inputs. This contradicts the assumption that $B(x)$ cannot, with high probability, be approximately computed classically. Since $\ell_\theta(O)$ approximates $B(x)$, as stated in Eq. (B2), it follows that $\ell_\theta(O)$ has a large variance over $\mathcal{P}_{\text{disc}}$. That is,

$$\mathcal{C}_{\text{Bool}}^{\text{disc}} \subset \overline{\text{BP}}. \quad (\text{B5})$$

Thus, $\mathcal{C}_{\text{Bool}}^{\text{disc}}$ provides an example of a problem class that is in $\overline{\text{BP}}$ but not in QESIM, as claimed.

However, this example is rather contrived. For one, it considers an initialization over a discrete set of parameters. This is a long way from the spirit of more variational quantum algorithms. Moreover, this discrete initialization is doing much of the heavy lifting in this example. To see this one can also consider a variant of the above problem where the parameters are sampled uniformly in the range $[0, 2\pi]$. We will call this distribution $\mathcal{P}_{\text{cont}}$ and the corresponding problem class $\mathcal{C}_{\text{Bool}}^{\text{cont}}$.

To understand both the absence of barren plateaus and classical simulability of $\mathcal{C}_{\text{Bool}}^{\text{cont}}$ it is helpful to write the backwards propagated observable in Eq. (B3), $A \equiv U^\dagger Z_1 U$, in terms of its local and global components. Specifically, for a small number k , we can split A as follows:

$$A = \underbrace{\sum_{|P| \leq k} \alpha_P P}_{A^L} + \underbrace{\sum_{|P| > k} \alpha_P P}_{A^H}. \quad (\text{B6})$$

We note that the expectation of A^L in any quantum state can be measured efficiently on quantum hardware in an initial data acquisition phase.

Next we highlight that for $\theta \sim \mathcal{P}_{\text{cont.}}$, the contribution of A^H to the loss decays exponentially with the bodyness k [86]. (Intuitively, this can broadly be understood as following from the fact that the expectation of any Pauli operator P of bodyness k in a typical random product state decays exponentially in k .) Hence, it follows that with high probability for $\theta \in \mathcal{P}_{\text{cont.}}$ the problem lives in an effective polynomial subspace and we have

$$\mathcal{C}_{\text{Bool}}^{\text{cont.}} \subset \mathcal{C}_{\text{polySub}}. \quad (\text{B7})$$

It then follows from Claim 2 in the main text that $\mathcal{C}_{\text{Bool}}^{\text{cont.}}$ is classically simulable in the sense that with high probability for $\theta \sim \mathcal{P}_{\text{cont.}}$ one can provide a classical estimate of the loss defined in Eq. (B3).

So what do we conclude from these examples? Firstly, this is a concrete example of the limitations of an average case notion of simulability. The loss we have constructed here is easy to classically simulate *on average* over the continuous parameter range $\mathcal{P}_{\text{cont.}}$ after an initial data acquisition phase. However, on a special measure-zero set of parameters, the problem becomes classically hard. Secondly, while in a typical random product state the expectation of any given global operator vanishes with high probability, something subtle can happen for the case of random computational basis states. Namely, while on average the contribution of a typical global Pauli is exponentially small, there are exponentially many such terms and with the right structure these can add up to give a non-zero total contribution. This is what happens for $\mathcal{C}_{\text{Bool}}^{\text{dist.}}$ —the contribution from A^H does not vanish and gives rise to a non-vanishing variance. However, in parallel, the exponential number of global terms in A^H makes its contribution impossible to classically simulate. This suggests smart initialization procedures may be the key to constructing efficiently trainable variational quantum algorithms that are not also classically simulable.

Supp. Info. C: Absence of barren plateaus in shallow local circuits with arbitrary topology

In this section we will show that that shallow hardware efficient ansatz composed of two-qubit gates acting an arbitrary topology do not posses barren plateaus. While these results can be, in a way, derived from those in [57, 58], we add the full proof here for completeness.

We begin by present a few useful definitions. First, let $\mathcal{H} = (\mathbb{C}^2)^{\otimes 2}$ be the Hilbert space of two qubits. In what follows we will denote the subsystem of the first qubit with an index A , and that of the second qubit with an index B so that $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. Then let $\mathcal{B}(\mathcal{H}^{\otimes t})$ be the set of bounded operators acting on t copies of the aforementioned Hilbert space. We are interested in computing expectation values of the form

$$\tau_U^{(t)}(X) = \int_{SU(4)} d\mu(U) U^{\otimes t} X (U^\dagger)^{\otimes t}, \quad (\text{C1})$$

where $d\mu(U)$ denotes the uniquely defined, normalized, Haar measure over the fundamental representation of $SU(4)$, and where $X \in \mathcal{B}(\mathcal{H}^{\otimes t})$.

We know that the twirls of Eq. (C1) can be evaluated via the Weingarten calculus [87], as $\tau^{(t)}(X)_U$ projects into the commutant of the t -th fold tensor representation of $SU(4)$. Here, we can use the Schur-Weyl duality to find that a basis of the commutant is the subsystem-permuting representation of S_t , the symmetric group of order t . For instance, for $t = 1$, a basis of S_1 is $\{\mathbb{1}\}$ where $\mathbb{1}$ denotes the 4×4 identity acting on $\mathcal{H}$. Hence, one finds

$$\tau_U^1(X) = \int_{SU(4)} d\mu(U) U X U^\dagger = \frac{\text{Tr}[X]}{4} \mathbb{1}. \quad (\text{C2})$$

Going further to $t = 2$, it is standard to take a basis of S_2 given by $\{\mathbb{1} \otimes \mathbb{1}, \text{SWAP}\}$. Here, SWAP denotes the operator that permutes states acting on each of the copies of the Hilbert space and whose action is given by

$$\text{SWAP}(|\psi_1\rangle \otimes |\psi_2\rangle) = |\psi_2\rangle \otimes |\psi_1\rangle, \quad (\text{C3})$$

for $|\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$. This leads to the standard result

$$\tau_U^2(X) = \frac{1}{15} \left(\text{Tr}[X] - \frac{\text{Tr}[X\text{SWAP}]}{4} \right) \mathbb{1} \otimes \mathbb{1} + \frac{1}{15} \left(\text{Tr}[X\text{SWAP}] - \frac{\text{Tr}[X]}{4} \right) \text{SWAP}. \quad (\text{C4})$$

While the previous form can, and has [88–92], been used to study properties of random quantum circuits composed of local gates, it has the key disadvantage that the basis $\{\mathbb{1} \otimes \mathbb{1}, \text{SWAP}\}$ is not orthogonal. As such, we will determine a more convenient basis to work in. For this purpose, we first re-arrange the Hilbert spaces so that

$$\mathcal{H}^{\otimes 2} = (\mathcal{H}_{A,1} \otimes \mathcal{H}_{B,1}) \otimes (\mathcal{H}_{A,2} \otimes \mathcal{H}_{B,2}) \rightarrow (\mathcal{H}_{A,1} \otimes \mathcal{H}_{A,2}) \otimes (\mathcal{H}_{B,1} \otimes \mathcal{H}_{B,2}), \quad (\text{C5})$$

so that

$$\mathbb{1} \otimes \mathbb{1} \rightarrow \mathbb{1}_A \otimes \mathbb{1}_B, \quad \text{SWAP} \rightarrow \text{SWAP}_A \otimes \text{SWAP}_B, \quad (\text{C6})$$

where $\text{SWAP}_{A(B)}$ are the operators that swap the $A(B)$ qubits in the two-fold tensor product of $\mathcal{H}$, and $\mathbb{1}_{A(B)}$ the identity operator on the two-copies of the subsystem's Hilbert space $\mathcal{H}_{A(B)}$. Moreover, let us write

$$\text{SWAP}_A = \frac{1}{2} \left(\mathbb{1}_{A,1} \otimes \mathbb{1}_{A,2} + X_{A,1} \otimes X_{A,2} + Y_{A,1} \otimes Y_{A,2} + Z_{A,1} \otimes Z_{A,2} \right) = \frac{1}{2} (\mathbb{1}_A + \text{SW}_A),$$

where we defined

$$\mathbb{1}_A = \mathbb{1}_{A,1} \otimes \mathbb{1}_{A,2}, \quad \text{SW}_A = X_{A,1} \otimes X_{A,2} + Y_{A,1} \otimes Y_{A,2} + Z_{A,1} \otimes Z_{A,2}. \quad (\text{C7})$$

Combining the previous results, we obtain

$$\text{SWAP} = \frac{1}{4} \left(\mathbb{1}_A \otimes \mathbb{1}_B + \text{SW}_A \otimes \mathbb{1}_B + \mathbb{1}_A \otimes \text{SW}_B + \text{SW}_A \otimes \text{SW}_B \right). \quad (\text{C8})$$

From here, we can find that an orthonormal basis for the commutant of the two-fold tensor rep of $SU(4)$ is

$$\left\{ \frac{\mathbb{1}_A \otimes \mathbb{1}_B}{4}, \frac{\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B}{4\sqrt{15}} \right\},$$

where we have dropped the trivial identity terms for ease of notation. Then, we can write the two-fold twirl as

$$\tau_U^2(X) = \frac{1}{16} \text{Tr}[X] \mathbb{1}_A \otimes \mathbb{1}_B + \frac{1}{240} \text{Tr}[X(\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B)] (\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B). \quad (\text{C9})$$

Given Eq. (C9) we can now compute the following twirls:

$$\tau_U^2(P_A \otimes P_B) = \frac{1}{15} (\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B), \quad (\text{C10})$$

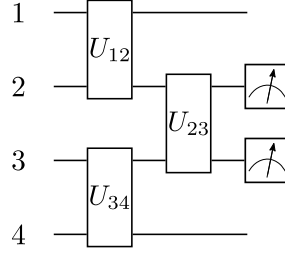
$$\tau_U^2(\text{SW}_A) = \frac{1}{5} (\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B), \quad (\text{C11})$$

$$\tau_U^2(\text{SW}_B) = \frac{1}{5} (\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B), \quad (\text{C12})$$

$$\tau_U^2(\text{SW}_A \otimes \text{SW}_B) = \frac{3}{5} (\text{SW}_A + \text{SW}_B + \text{SW}_A \otimes \text{SW}_B), \quad (\text{C13})$$

where $P_{A(B)}$ is a single-qubit Pauli acting on the subsystems $A(B)$.

Using the previous equation, we can evaluate the loss function variance over arbitrary circuits composed of $SU(4)$ gates by simply keeping track of how the SW operators transform between themselves and how the coefficients accumulate. As an example, consider the four qubit circuit shown in Fig. 1, where the unitary U implementing the whole circuit can be expressed as $U = U_{23}U_{34}U_{12}$. We assume the input to the circuit is ρ , and that at the end of the circuit we measure the



Supplementary Figure 1. **Four qubit circuit example.** Consider the following circuit acting on four qubits. At the end of the circuit we measure a Pauli operator on qubits 2 and 3.

expectation value of some Pauli operator acting on qubits 2 and 3. We further assume that all the two-qubit gates form independent two-design over $SU(4)$. We want to compute a quantities of the for $\mathbb{E}_U[\text{Tr}[U\rho U^\dagger O]]$ and $\mathbb{E}_U[\text{Tr}[U\rho U^\dagger O]^2]$. In particular we will evaluate the expectation value over the circuit as

$$\mathbb{E}_U[f(U)] = \int_{SU(4)} d\mu(U_{12})d\mu(U_{34})d\mu(U_{23})f(U), \quad (\text{C14})$$

Using Eq. (C2), we can readily show that

$$\mathbb{E}_U[\text{Tr}[U\rho U^\dagger O]] = 0. \quad (\text{C15})$$

Then, since

$$\begin{aligned} \text{Tr}[U\rho U^\dagger O]^2 &= \text{Tr}[U^{\otimes 2}\rho^{\otimes 2}(U^\dagger)^{\otimes 2}O] \\ &= \text{Tr}[\rho^{\otimes 2}(U^\dagger)^{\otimes 2}OU^{\otimes 2}], \end{aligned}$$

we find

$$\mathbb{E}_U[\text{Tr}[\rho^{\otimes 2}(U^\dagger)^{\otimes 2}OU^{\otimes 2}]] = \text{Tr}[\rho^{\otimes 2}\mathbb{E}_U[(U^\dagger)^{\otimes 2}OU^{\otimes 2}]],$$

where we have used the linearity of the trace. Then, we see that we need to compute

$$\tau_{U_{12}}^{(2)} \circ \tau_{U_{34}}^{(2)} \circ \tau_{U_{23}}^{(2)}[O^{\otimes 2}]. \quad (\text{C16})$$

Since O is some Pauli acting non trivially on qubits 2 and 3, we find via Eq. (C10)

$$\tau_{U_{23}}^{(2)}[O^{\otimes 2}] = \frac{1}{15}(\text{SW}_2 + \text{SW}_3 + \text{SW}_2 \otimes \text{SW}_3). \quad (\text{C17})$$

We then use Eq. (C11)–(C13) to obtain

$$\begin{aligned} &\tau_{U_{34}}^{(2)}\left[\frac{1}{15}(\text{SW}_2 + \text{SW}_3 + \text{SW}_2 \otimes \text{SW}_3)\right] \\ &= \frac{\text{SW}_2}{15} + \frac{1}{75}(\text{SW}_3 + \text{SW}_4 + \text{SW}_3 \otimes \text{SW}_4) + \frac{3}{75}(\text{SW}_2 \otimes \text{SW}_3 + \text{SW}_2 \otimes \text{SW}_4 + \text{SW}_2 \otimes \text{SW}_3 \otimes \text{SW}_4). \end{aligned} \quad (\text{C18})$$

Here, we can already observe an important phenomenon. If a SW_i goes into a twirl from a gate U_{ij} , then we obtain terms SW_i , SW_j , and $\text{SW}_i \otimes \text{SW}_j$ with a coefficient $\frac{1}{5}$. Similarly, if $\text{SW}_i \otimes \text{SW}_j$ goes into a twirl from a gate U_{ij} , then we obtain terms SW_i , SW_j , and $\text{SW}_i \otimes \text{SW}_j$ with a coefficient $\frac{3}{5}$. Hence, we simply need to keep track of indexes and coefficients. As such, let us we-write Eq. (C17) as

$$O^{\otimes 2} \xrightarrow{\tau_{U_{23}}^{(2)}} \left\{ \left\{ (2), \frac{1}{5} \right\}, \left\{ (3), \frac{1}{5} \right\}, \left\{ (2, 3), \frac{1}{5} \right\} \right\}. \quad (\text{C19})$$

Then, the next twirl over U_{34} in Eq. (C18) leads to

$$\xrightarrow{\tau_{U_{33}}^{(2)}} \left\{ \left\{ (2), \frac{1}{5} \right\}, \left\{ (3), \frac{1}{75} \right\}, \left\{ (4), \frac{1}{75} \right\}, \left\{ (3, 4), \frac{1}{75} \right\}, \right. \\ \left. \left\{ (2, 3), \frac{3}{75} \right\}, \left\{ (2, 4), \frac{3}{75} \right\}, \left\{ (2, 3, 4), \frac{3}{75} \right\} \right\} \quad (\text{C20})$$

and finally the next twirl over U_{12}

$$\xrightarrow{\tau_{U_{33}}^{(2)}} \left\{ \left\{ (1), \frac{1}{75} \right\}, \left\{ (2), \frac{1}{75} \right\}, \left\{ (3), \frac{1}{75} \right\}, \left\{ (4), \frac{1}{75} \right\}, \right. \\ \left\{ (1, 2), \frac{1}{75} \right\}, \left\{ (3, 4), \frac{1}{75} \right\}, \left\{ (2, 3), \frac{3}{375} \right\}, \\ \left\{ (1, 3), \frac{3}{375} \right\}, \left\{ (2, 4), \frac{3}{375} \right\}, \left\{ (1, 4), \frac{3}{375} \right\}, \\ \left\{ (1, 2, 3), \frac{3}{375} \right\}, \left\{ (1, 2, 4), \frac{3}{375} \right\}, \left\{ (2, 3, 4), \frac{3}{375} \right\}, \\ \left. \left\{ (1, 3, 4), \frac{3}{375} \right\}, \left\{ (1, 2, 3, 4), \frac{3}{375} \right\} \right\}. \quad (\text{C21})$$

The previous example unveils a simple pattern that will be universally true. First, the “most-local” operators, are the ones which will be suppressed the least. Whereas “more global” operators will be quickly suppressed. As such, we can intuitively see that, in average, the Heisenberg evolved operator will live on $\mathcal{O}(1)$ bodied operators. Indeed, one can study how much the coefficient of a given operators at the out put of the circuit will look like by evaluating some weighted random walk over the circuit architecture following the rules mentioned above.

Putting it all together, we are now ready to show that shallow local circuits avoid barren plateaus when measured locally at their output. Interestingly, this result encompassed, and constitutes a simpler proof, to those in Refs. [40] and [78]. Consider therefore an arbitrary circuit with two-qubit gate topology

$$\mathcal{A} = \{(i_k, j_k), \quad \forall k = 1, \dots, K\}. \quad (\text{C22})$$

Note that we are not assuming any dimensionality on $\mathcal{A}$, meaning that it could be as general as desired. Then, if the measurement operator is locally acting on the μ -th qubit, and that this index appears in $\mathcal{A}$ L times, one can lower bound the expectation value’s variance as

$$\text{Var}_U[\text{Tr}[U\rho U^\dagger O]] \geq \frac{1}{15} \frac{1}{5^{L-1}} \text{Tr}[\rho^{\otimes 2} \text{SW}_\mu] \\ = \frac{1}{15} \frac{2}{5^{L-1}} (\text{Tr}[\rho_\mu^2] - \frac{1}{2}), \quad (\text{C23})$$

where we have defined as ρ_μ^2 the reduced state of ρ on the μ -th qubit, and where we have used the fact that $\text{SW}_\mu = 2(\text{SWAP}_\mu - \mathbb{1})$. Moreover, here we have used the fact that we can drop all other terms in the path summation since the overlap between $\rho^{\otimes 2}$ and the tensor product of many SW’s is always positive. More generally, if we define as $\mathcal{S}$ as the set of indexes which are connected through a path of length L from the beginning of the circuit to the measurement on the μ -th qubit, one can tighten the bound as

$$\text{Var}_U[\text{Tr}[U\rho U^\dagger O]] \geq \frac{1}{15} \frac{2}{5^{L-1}} \sum_{j \in \mathcal{S}} (\text{Tr}[\rho_j^2] - \frac{1}{2}). \quad (\text{C24})$$

The previous result shows that if the circuit is shallow, i.e., if there exists a path of length L such that $L \in \mathcal{O}(\log(n))$; and if the input state ρ is not too entangled, i.e., $|\text{Tr}[\rho_j^2] - \frac{1}{2}| \in \Omega(1/\text{poly}(n))$ then

$$\text{Var}_U[\text{Tr}[U\rho U^\dagger O]] \in \Omega\left(\frac{1}{\text{poly}(n)}\right). \quad (\text{C25})$$

Notably, we find that the reduced states will be exponentially close to the maximally mixed state if they satisfy a volume law of entanglement, whereas their distance to it will be polynomially small if they follow an area law of entanglement [93].

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